Below is a foundational PRD structure for what could realistically become an enterprise-scale category-defining platform.

This is intentionally written as:

- infrastructure-first
- modular
- cloud-native
- sovereign-ready
- multi-tenant
- agent-runtime-centric

rather than another "AI SaaS app."

You are not building a chatbot company.

You are building:

Agentic Operating Infrastructure

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Project Codename

ORCHESTRA

Agentic Department Operating System (ADOS)

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1. EXECUTIVE SUMMARY

ORCHESTRA is a multi-tenant enterprise platform that enables organizations to deploy fully operational AI-native departments on demand using coordinated agent swarms, institutional memory systems, workflow orchestration, and governance-driven execution frameworks.

Organizations can:

- select a department template
- ingest institutional knowledge
- configure governance and policies
- deploy operational agent teams
- supervise execution
- scale departments elastically

The platform transforms AI from:

- assistant tooling

into:

- operational organizational infrastructure.

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2. PRODUCT VISION

To become the foundational operating system for AI-native enterprises and governments by enabling deployable digital departments that operate with institutional memory, governance, explainability, and sovereign-grade control.

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3. CORE PRODUCT THESIS

Organizations do not fundamentally require:

- more employees
- more dashboards
- more copilots

They require:

- scalable operational cognition
- persistent institutional memory
- reliable process execution
- governed autonomous systems

ORCHESTRA provides this through:

- agentic departments
- modular swarm architectures

- organizational memory systems
- workflow governance
- human-supervised autonomy

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4. PLATFORM PRINCIPLES

4.1 Sovereign by Design

- full on-prem support
- regional deployment isolation
- air-gapped deployment compatibility
- private model hosting
- private vector stores
- enterprise encryption

4.2 Multi-Tenant Isolation

Every tenant receives:

- isolated memory
- isolated orchestration
- isolated models
- isolated governance
- isolated execution logs
- isolated identity layer

4.3 Human-Governed Autonomy

Agents may:

- recommend
- analyze
- coordinate
- draft
- execute limited actions

Critical actions require:

- approval workflows
- escalation paths
- policy validation

4.4 Explainability First

Every action must support:

- traceability
- replayability
- provenance
- source attribution
- confidence scoring
- execution auditability

4.5 Modular Runtime Architecture

All major platform systems must be independently deployable and replaceable.

No hard vendor lock-in.

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5. HIGH-LEVEL SYSTEM ARCHITECTURE

```
┌─────────────────────────────────────────────┐
│              EXPERIENCE LAYER               │
├─────────────────────────────────────────────┤
│ Web UI │ API │ SDK │ CLI │ Mobile           │
└─────────────────────────────────────────────┘

┌─────────────────────────────────────────────┐
│            ORCHESTRATION LAYER              │
├─────────────────────────────────────────────┤
│ Agent Runtime │ Workflow Engine             │
│ Scheduling │ Delegation │ Coordination      │
└─────────────────────────────────────────────┘

┌─────────────────────────────────────────────┐
│               MEMORY LAYER                  │
├─────────────────────────────────────────────┤
```

```
│ Vector DB │ Graph Memory │ Episodic         │
│ Semantic Memory │ Knowledge Graph           │
└───────────────────────────────────────────────┘

┌───────────────────────────────────────────────┐
│                EXECUTION LAYER                │
├───────────────────────────────────────────────┤
│ Tool Calling │ MCP │ Connectors               │
│ APIs │ External Systems │ Actions             │
└───────────────────────────────────────────────┘

┌───────────────────────────────────────────────┐
│               GOVERNANCE LAYER                │
├───────────────────────────────────────────────┤
│ Policies │ RBAC │ Approval Chains             │
│ Guardrails │ Audit │ Compliance               │
└───────────────────────────────────────────────┘

┌───────────────────────────────────────────────┐
│             INFRASTRUCTURE LAYER              │
├───────────────────────────────────────────────┤
│ Kubernetes │ GPU Runtime │ Queues             │
│ Storage │ Observability │ Security            │
└───────────────────────────────────────────────┘
```

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6. MODULAR ARCHITECTURE

MODULE 01 — TENANT CONTROL PLANE

Purpose

Centralized tenant provisioning and lifecycle management.

Responsibilities

- tenant onboarding
- namespace creation
- quota allocation
- billing integration
- deployment lifecycle
- environment isolation

Components

- tenant registry
- tenant policy engine
- deployment manager
- subscription manager
- tenant secrets manager

APIs

```
POST /tenants
GET /tenants/{id}
PATCH /tenants/{id}
DELETE /tenants/{id}
```

MODULE 02 — IDENTITY & ACCESS MANAGEMENT

Purpose

Enterprise-grade authentication and authorization.

Capabilities

- SSO
- LDAP
- Active Directory
- SCIM
- RBAC
- ABAC
- service identities
- agent identities

Required Features

- MFA
- audit trails
- session replay
- delegated admin roles

Suggested Stack

- Keycloak
- Ory

- Authentik

MODULE 03 — AGENT ORCHESTRATION ENGINE

Purpose

Core runtime engine for agent swarms.

Responsibilities

- workflow execution
- agent scheduling
- state management
- task routing
- delegation
- retry logic
- timeout handling
- escalation logic

Runtime Model

Event-driven graph orchestration.

Required Features

- DAG execution
- async execution
- resumable workflows
- distributed execution
- state checkpointing
- workflow replay

Suggested Stack

- LangGraph
- Temporal
- Ray
- Celery/Kafka

MODULE 04 — AGENT REGISTRY

Purpose

Central registry for reusable agents.

Capabilities

- agent versioning
- capability indexing
- permissions
- dependency management
- runtime constraints
- cost profiles

Agent Metadata

```
agent:
  name:
  role:
  capabilities:
  tools:
  memory_access:
  escalation_rules:
  governance_policies:
```

MODULE 05 — DEPARTMENT TEMPLATE ENGINE

Purpose

Deploy reusable organizational structures.

Core Concept

A department is:

- agents
- workflows

- memory topology
- governance rules
- KPIs
- integrations
- operational policies

Example Templates

- HR
- Finance
- Procurement
- Engineering
- Translation
- Legal
- Compliance
- Research

Deployment Flow

```
Select Template
→ Configure Policies
→ Connect Data Sources
→ Provision Memory
→ Deploy Swarm
→ Begin Operations
```

MODULE 06 — KNOWLEDGE INGESTION PIPELINE

Purpose

Transform enterprise data into operational memory.

Data Sources

- PDFs
- SharePoint
- ERP systems
- email
- databases
- APIs
- Confluence

- Jira
- CRMs
- document repositories

Pipeline Stages

```
Ingestion
→ OCR
→ Parsing
→ Chunking
→ Classification
→ Ontology Mapping
→ Entity Extraction
→ Embedding
→ Indexing
→ Graph Construction
```

Required Features

- multilingual ingestion
- Arabic-native NLP
- metadata extraction
- semantic deduplication
- version tracking
- lineage mapping

MODULE 07 — MEMORY FABRIC

Purpose

Persistent organizational cognition layer.

Memory Types

Semantic Memory

Facts and embeddings.

Episodic Memory

Historical executions.

Procedural Memory

Operational workflows.

Graph Memory

Relationship structures.

Institutional Memory

Long-term organizational intelligence.

Suggested Stack

- Qdrant
- Neo4j
- PostgreSQL
- Redis

MODULE 08 — TOOL EXECUTION LAYER

Purpose

Secure external action execution.

Capabilities

- API execution
- browser automation
- ERP interaction
- email sending
- document generation
- workflow triggering

Standards

- MCP-native
- tool sandboxing
- capability isolation

Required Features

- execution permissions
- policy validation
- dry-run mode
- rollback support

MODULE 09 — HUMAN SUPERVISION LAYER

Purpose

Govern autonomous execution safely.

Core Functions

- approvals
- escalations
- intervention
- override controls
- human-in-the-loop review

Approval Models

- sequential approvals
- parallel approvals
- conditional approvals
- threshold approvals

MODULE 10 — POLICY & GOVERNANCE ENGINE

Purpose

Enterprise compliance and control.

Capabilities

- policy enforcement
- action validation
- data governance
- compliance auditing
- risk scoring

Policy Examples

```
Finance agents cannot approve payments above threshold.

HR agents cannot access executive compensation data.

Legal agents require human approval before contract execution.
```

Suggested Stack

- Open Policy Agent
- Rego

MODULE 11 — OBSERVABILITY & TELEMETRY

Purpose

Operational transparency.

Metrics

- token usage
- workflow latency
- agent reliability
- hallucination rates
- execution cost
- task completion
- escalation frequency

Capabilities

- distributed tracing
- replay systems
- execution graphs
- anomaly detection
- cost forecasting

Suggested Stack

- OpenTelemetry
- Grafana
- Prometheus
- Loki

MODULE 12 — MODEL ABSTRACTION LAYER

Purpose

Decouple orchestration from model vendors.

Supported Models

- OpenAI
- Anthropic
- Gemini
- local LLMs
- sovereign Arabic models

Features

- routing
- fallback models
- cost optimization
- latency optimization
- capability scoring

MODULE 13 — MULTI-MODEL ROUTING ENGINE

Purpose

Intelligent model assignment.

Example Logic

```
Translation tasks
→ Arabic sovereign model

Reasoning tasks
→ frontier reasoning model

Low-cost summaries
→ small local model
```

Features

- dynamic routing
- token optimization
- quality scoring
- energy-aware scheduling

MODULE 14 — AGENT COMMUNICATION BUS

Purpose

Inter-agent messaging infrastructure.

Features

- async messaging
- event streaming
- publish/subscribe
- workflow signals
- state synchronization

Suggested Stack

- Kafka
- NATS
- RabbitMQ

MODULE 15 — AGENT MARKETPLACE

Purpose

Third-party ecosystem.

Marketplace Assets

- agents
- workflows
- department templates
- connectors
- governance packs
- compliance packs

Revenue Model

- revenue sharing
- subscriptions
- usage licensing

MODULE 16 — EXECUTION SANDBOX

Purpose

Secure runtime isolation.

Required Features

- container isolation
- syscall restrictions

- ephemeral execution
- resource quotas
- network restrictions

MODULE 17 — COST GOVERNANCE ENGINE

Purpose

Prevent runaway operational costs.

Features

- token budgets
- execution quotas
- agent throttling
- usage forecasting
- tenant billing

MODULE 18 — ENTERPRISE CONNECTOR FABRIC

Integrations

- SAP
- Oracle
- Salesforce
- Microsoft 365
- Google Workspace
- ServiceNow
- Slack
- Teams
- Jira
- GitHub

MODULE 19 — ARABIC LANGUAGE INTELLIGENCE LAYER

Strategic ADRI Differentiator

Features

- Arabic-first embeddings
- terminology governance
- dialect normalization
- Arabic OCR
- Arabic semantic retrieval
- translation memory
- multilingual reasoning
- Arabic plagiarism intelligence

This module alone can become a national strategic moat.

MODULE 20 — SOVEREIGN DEPLOYMENT FABRIC

Deployment Targets

- public cloud
- private cloud
- air-gapped
- sovereign national infrastructure
- edge deployment

KSA Requirements

- local hosting
- data residency
- compliance enforcement
- sovereign AI compatibility

---

7. DEPARTMENT OBJECT MODEL

```
department:
  id:
  name:
  purpose:
  governance_policy:
  agents:
  workflows:
  memory_topology:
  escalation_rules:
  integrations:
  observability_profile:
```

---

8. AGENT OBJECT MODEL

```
agent:
  id:
  role:
  objectives:
  permissions:
  tools:
  memory_scope:
  policies:
  escalation_targets:
  execution_budget:
```

---

9. MULTI-TENANT ISOLATION MODEL

Isolation Levels

Logical Isolation

Namespace separation.

Data Isolation

Separate storage.

Cryptographic Isolation

Tenant-specific encryption keys.

Execution Isolation

Dedicated runtime environments.

---

10. SECURITY REQUIREMENTS

Mandatory

- zero trust
- encrypted memory
- encrypted vector stores
- RBAC/ABAC
- execution signing
- audit trails
- immutable logs
- secret rotation

---

11. SCALABILITY TARGETS

Initial

- 100 tenants
- 10,000 agents
- 1M workflow executions/day

Long-Term

- national-scale deployments
- 100M executions/day
- federated agent ecosystems

---

12. BUSINESS MODEL

SaaS

Per-seat + execution-based.

Enterprise

Private deployment licensing.

Sovereign

National infrastructure licensing.

Marketplace

Revenue share.

---

13. STRATEGIC MOATS

Technical Moats

- orchestration reliability
- institutional memory
- governance infrastructure
- Arabic AI specialization

Business Moats

- department templates
- workflow intelligence
- compliance packs
- ecosystem marketplace

---

14. CRITICAL RISKS

Technical

- hallucinated execution
- workflow instability
- memory poisoning
- infinite loops
- runaway costs

Operational

- enterprise trust
- compliance failures
- governance gaps

Strategic

- foundation model commoditization

---

15. THE MOST IMPORTANT STRATEGIC DECISION

Do NOT position this as:

- “AI assistant platform”
- “multi-agent platform”
- “workflow automation”

Position this as:

“Enterprise Agentic Workforce Infrastructure”

That changes:

- pricing
- perception
- enterprise valuation
- procurement category
- competitive landscape

You stop competing with AI startups.

You start competing with:

- BPO firms
- ERP vendors
- enterprise operating systems

- consulting firms
- workforce infrastructure providers

That is a much larger strategic category.